
AR 515: Sensors and State Estimation

Course Code	AR 515
Course Name	Sensors and State Estimation
L-T-P-C	3-0-0-3
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Sensors:** Introduction and motivation, different types of sensors and their real time applications, signal conditioning, classic vs smart sensors. (10 hours)
- **Recursive State Estimation:** Overview, basic concept in probability, robot environment interaction, Bayes filter. (7 hours)
- **Filtering Techniques:** Introduction, Kalman Filter (KF), Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), Particle Filter, and Quantum Stochastic Filtering. (15 hours)
- **Multi-sensor Data Fusion:** Introduction to multi-sensor systems, some examples like Unmanned Aircraft System (UAS), reference frame for multi-sensor fusion, calibration, synchronization, multi-sensor fusion with EKF. Case study and course projects. (10 hours)

Text Books

1. Thrun, Sebastian, Probabilistic robotics, **Communications of the ACM** 45.3: 52-57, 2002.
2. Fraden, Jacob., **Handbook of modern sensors: physics, designs, and applications**, 357-359, 1998.

References

1. Bhuyan, Manabendra, **Intelligent Instrumentation: Principles and Applications**, CRC Press, 2010.
2. Sawhney, A. K., **Electrical and electronic Measurements and Instrumentation**, 1985.
3. Behera, Laxmidhar, and Indrani Kar, Quantum stochastic filtering, 2005 **IEEE International Conference on Systems, Man and Cybernetics**, Vol. 3. IEEE, 2005.
4. Meijer, Gerard, (ed.), **Smart sensor systems**, John Wiley & Sons, 2008.
5. Jacob Fraden, **Handbook of modern Sensors**, AIP Press, 1997.
6. E. O. Deobelin and D. Manik, **Measurement Systems – Application and Design**, Tata McGraw-Hill, 2004.
7. Yallup, Kevin, and Krzysztof Iniewski, (eds.), **Technologies for smart sensors and sensor fusion**, CRC Press, 2014.
8. Prosser, Stephen J., and Ernest DD Schmidt, **Smart sensors for industrial applications**, Sensor Review, 1997.